

Unit 1: Basic Principle of Counting, Permutations and Combinations

Short Answer Questions with Solutions

1. How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7?

- (a) **Repetition not allowed:** Number of choices for first digit = 5 (any of the 5 digits), for second digit = 4 (remaining), for third digit = 3 (remaining).

$$\text{Total} = 5 \times 4 \times 3 = 60$$

- (b) **Repetition allowed:** Number of choices for each place = 5, so

$$\text{Total} = 5 \times 5 \times 5 = 125$$

2. 4 boys and 3 girls seating arrangement:

- (a) **All boys sit together:** Treat 4 boys as one block. Then we have (1 block of boys + 3 girls) = 4 objects. Number of arrangements of these = 4!. Boys within block can be arranged in 4! ways.

$$\text{Total} = 4! \times 4! = 24 \times 24 = 576$$

- (b) **Boys and girls sit alternately:** Start with boys: B G B G B G B Boys can be arranged in 4! ways, girls in 3! ways.

$$\text{Total} = 4! \times 3! = 24 \times 6 = 144$$

3. **4-letter words from NEPAL:** The word NEPAL has 5 distinct letters. Number of arrangements of 4 letters from 5 distinct = ${}^5P_4 = \frac{5!}{(5-4)!} = 5 \times 4 \times 3 \times 2 = 120$.

4. **Arrangements of MANAGEMENT:** Letters: M A N A G E M E N T Frequencies: M(2), A(2), N(2), E(2), G(1), T(1) → Total letters = 10.

$$\text{Total} = \frac{10!}{2! \times 2! \times 2! \times 2!} = \frac{3628800}{16} = 226800$$

5. Committee from 7 men and 5 women:

- (a) Exactly 2 women: Choose 2 women from 5 and 2 men from 7:

$${}^5C_2 \times {}^7C_2 = 10 \times 21 = 210$$

- (b) At least 1 woman: Total committees ${}^{12}C_4 = 495$. Committees with no woman = ${}^7C_4 = 35$.

$$\text{Required} = 495 - 35 = 460$$

Long Answer Questions with Solutions

6. Group of 5 students from 5 boys and 4 girls:

- (a) **At least 2 girls:**

Girls	Boys	Number of ways to choose	Total ways
2	3	${}^4C_2 \times {}^5C_3 = 6 \times 10$	60
3	2	${}^4C_3 \times {}^5C_2 = 4 \times 10$	40
4	1	${}^4C_4 \times {}^5C_1 = 1 \times 5$	5
Total			105

- (b) **No restriction:**

$$\text{Choose any 5 from 9 students: } {}^9C_5 = 126$$

7. **Arrangements of PROBABILITY with vowels together:** Vowels = O, A, I, I (4 vowels, I repeats twice). Treat vowels as one block + consonants (P, R, B, B, L, T, Y). Consonants count: 7 letters with B repeated twice. Arrangements of consonants = $\frac{7!}{2!} = 2520$. Vowels arrangement = $\frac{4!}{2!} = 12$.

$$\text{Total} = 2520 \times 12 = 30240$$

8. **From the word MATHS:** All the letters M, A, T, H, S are distinct.

Solution by Direct Counting:

- (a) **Number of 5-letter words beginning with M:** Since the word must begin with the letter M, we fix M in the first position. The remaining 4 positions can be filled with the remaining 4 letters (A, T, H, S) in any order. The number of ways to arrange these 4 letters is the number of permutations of 4 distinct letters:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

So, total number of 5-letter words beginning with M is $\boxed{24}$.

- (b) **Number of 5-letter words that do not begin with M:** Here, the first letter can be any of the other 4 letters (A, T, H, S). So, there are 4 choices for the first position. After choosing the first letter, we arrange the remaining 4 letters (including M and the other 3 letters) in the remaining 4 positions. The number of ways to arrange these 4 letters is $4!$. Therefore, total number of such words is:

$$4 \times 4! = 4 \times 24 = 96$$

Hence, total number of 5-letter words not beginning with M is $\boxed{96}$.

- (c) **Number of 5-letter words beginning with M and not ending with S:** Fix M in the first position. The last letter cannot be S, so it must be chosen from the other 3 letters (A, T, H). So, there are 3 choices for the last position. Now, the 3 middle positions are to be filled with the remaining 3 letters after fixing first and last letters. The number of ways to arrange these 3 letters is $3!$. Hence, total number of such words is:

$$3 \times 3! = 3 \times 6 = 18$$

So, total number of 5-letter words beginning with M and not ending with S is $\boxed{18}$.

or

Alternative method:

- (a) **Begins with M:** Total arrangements = 120. Number of words not starting with M = $4 \times 4! = 96$. Subtract:

$$120 - 96 = 24$$

- (b) **Does not begin with M:** Total arrangements = 120. Number of words starting with M = $4! = 24$. Subtract:

$$120 - 24 = 96$$

- (c) **Begins with M and does not end with S:** Total words starting with M = $4! = 24$. Words starting with M and ending with S = $3! = 6$. Subtract:

$$24 - 6 = 18$$

9. **Passwords with 2 letters + 3 digits:**

- (a) Repetition allowed: Letters: 26×26 , Digits: $10 \times 10 \times 10$

$$26^2 \times 10^3 = 676 \times 1000 = 676000$$

- (b) Repetition not allowed: Letters: ${}^{26}P_2 = 26 \times 25 = 650$, Digits: ${}^{10}P_3 = 10 \times 9 \times 8 = 720$

$$\text{Total} = 650 \times 720 = 468000$$

10. **5-digit even numbers from 1,2,3,4,5,6 without repetition:** Last digit options (even) = 2, 4, 6 $\rightarrow$ 3 choices. Remaining 4 digits can be arranged in ${}^5P_4 = 5 \times 4 \times 3 \times 2 = 120$ ways.

$$\text{Total} = 3 \times 120 = 360$$

11. Team of 6 students from 8 boys and 7 girls:

(a) At least 3 girls:

Possible combinations:

Girls	Boys	Number of ways to choose	Total ways
3	3	${}^7C_3 \times {}^8C_3 = 35 \times 56$	1960
4	2	${}^7C_4 \times {}^8C_2 = 35 \times 28$	980
5	1	${}^7C_5 \times {}^8C_1 = 21 \times 8$	168
6	0	${}^7C_6 \times {}^8C_0 = 7 \times 1$	7
Total			3115

(b) More girls than boys:

Possible combinations:

Girls	Boys	Number of ways to choose	Total ways
4	2	${}^7C_4 \times {}^8C_2 = 35 \times 28$	980
5	1	${}^7C_5 \times {}^8C_1 = 21 \times 8$	168
6	0	${}^7C_6 \times {}^8C_0 = 7 \times 1$	7
Total			1155

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Unit 1B: Permutations and Combinations

Short Answer Questions - Solutions

1. Evaluate: 7P_4 and 7C_4 .

$${}^7P_4 = \frac{7!}{(7-4)!} = \frac{7!}{3!} = \frac{5040}{6} = 840$$

$${}^7C_4 = \frac{7!}{4! \times 3!} = \frac{5040}{24 \times 6} = 35$$

2. If $nP_3 = 120$, find n .

$$nP_3 = \frac{n!}{(n-3)!} = 120$$

Try values:

$$5P_3 = \frac{5!}{2!} = \frac{120}{2} = 60 \quad (\text{too small})$$

$$6P_3 = \frac{6!}{3!} = \frac{720}{6} = 120 \quad \Rightarrow n = 6$$

3. Find the number of ways to arrange the letters of the word **NEPAL**.

Number of letters = 5 (all distinct)

Number of arrangements = $5! = 120$

4. In how many ways can 3 girls be selected from a group of 5?

$${}^5C_3 = \frac{5!}{3!2!} = 10$$

5. Prove that ${}^nC_r = {}^nC_{n-r}$.

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)!r!} = {}^nC_{n-r}$$

6. Find r if ${}^8C_r = {}^8C_{r+2}$.

Since ${}^nC_r = {}^nC_{n-r}$, we have:

$$r = 8 - (r + 2) \implies r = 6 - r \implies 2r = 6 \implies r = 3$$

7. Find the value of $\frac{{}^8P_3}{{}^8C_3}$.

$$\begin{aligned} {}^8P_3 &= \frac{8!}{5!} = 8 \times 7 \times 6 = 336 \\ {}^8C_3 &= \frac{8!}{3! \times 5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 \end{aligned}$$

Therefore,

$$\frac{{}^8P_3}{{}^8C_3} = \frac{336}{56} = 6$$

8. In how many ways can 3 people be chosen from 6?

$${}^6C_3 = \frac{6!}{3!3!} = 20$$

9. A committee of 3 men and 2 women is to be formed from 5 men and 4 women. In how many ways can this be done?

$${}^5C_3 \times {}^4C_2 = 10 \times 6 = 60$$

10. How many 3-digit even numbers can be formed using digits 2, 4, 6, 7, and 8 without repetition?

Last digit (units place) must be even: 2, 4, 6, 8 $\Rightarrow$ 4 choices

First digit: choose from remaining 4 digits (excluding the chosen last digit)

Second digit: choose from remaining 3 digits

$$\text{Total numbers} = 4 \times 4 \times 3 = 48$$

Long Answer Questions - Solutions

11. How many 5-digit numbers can be formed using digits 1–5 if:

- (a) **Repetition is not allowed:**

$$\text{Number of ways} = 5! = 120$$

- (b) **Repetition is allowed:**

$$\text{Each digit can be chosen independently: } 5^5 = 3125$$

12. Prove that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

Starting from definitions:

$${}^nC_r + {}^nC_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$$

Finding common denominator and simplifying leads to:

$${}^{n+1}C_r = \frac{(n+1)!}{r!((n+1)-r)!}$$

This is a standard combinatorial identity (Pascal's Rule).

13. In how many ways can the letters of the word **SUCCESS** be arranged?

Letters count: S=3, C=2, U=1, E=1

Total letters = 7

Number of arrangements:

$$\frac{7!}{3! \times 2! \times 1! \times 1!} = \frac{5040}{6 \times 2} = 420$$

14. From 6 men and 4 women, how many committees of 5 can be formed that include at least 2 women?

$$\text{Total committees} = {}^{10}C_5 = 252$$

$$\text{Committees with 0 women} = {}^6C_5 = 6$$

$$\text{Committees with 1 woman} = {}^4C_1 \times {}^6C_4 = 4 \times 15 = 60$$

$$\text{Committees with at least 2 women} = 252 - (6 + 60) = 186$$

15. If ${}^nP_2 = 56$, find n .

$${}^nP_2 = \frac{n!}{(n-2)!} = n(n-1) = 56$$

Solve quadratic:

$$n^2 - n - 56 = 0$$

$$n = \frac{1 \pm \sqrt{1 + 224}}{2} = \frac{1 \pm 15}{2}$$

Positive root:

$$n = \frac{1 + 15}{2} = 8$$

16. How many arrangements can be made of the letters of the word **MANAGEMENT** so that all vowels occur together?

Word: M A N A G E M E N T

Vowels: A, A, E, E (4 vowels, with A and E repeated twice)

Consonants: M, N, G, M, N, T (6 consonants, with M and N repeated twice each)

Treat all vowels as one entity, total entities = 6 consonants + 1 vowel group = 7

Number of ways to arrange 7 entities:

$$\frac{7!}{2! \times 2!} = \frac{5040}{4} = 1260$$

Number of ways to arrange vowels inside the group:

$$\frac{4!}{2! \times 2!} = \frac{24}{4} = 6$$

Total arrangements:

$$1260 \times 6 = 7560$$

17. From a group of 5 boys and 3 girls, how many teams of 4 can be formed with exactly 2 girls?

$$\text{Choose 2 girls from 3: } {}^3C_2 = 3$$

$$\text{Choose 2 boys from 5: } {}^5C_2 = 10$$

$$\text{Total teams} = 3 \times 10 = 30$$

18. Find the number of circular permutations of 6 boys around a round table.

Number of circular permutations of n distinct objects:

$$(n-1)! = 5! = 120$$

19. Find the number of 4-digit numbers greater than 5000 that can be formed using digits 2, 3, 5, 6, and 7.

First digit must be ≥ 5 (digits available: 5, 6, 7)

Number of choices for first digit = 3

Remaining digits: choose 3 digits from remaining 4 digits without repetition and arrange them

Number of ways for remaining 3 digits = $4 \times 3 \times 2 = 24$

Total numbers = $3 \times 24 = 72$

20. A committee of 4 is to be formed from 7 men and 5 women. How many ways are there to form a committee with at most 2 women?

Committees with 0 women:

$${}^7C_4 = 35$$

Committees with 1 woman:

$${}^5C_1 \times {}^7C_3 = 5 \times 35 = 175$$

Committees with 2 women:

$${}^5C_2 \times {}^7C_2 = 10 \times 21 = 210$$

Total committees with at most 2 women:

$$35 + 175 + 210 = 420$$